SPECTRAL METAMERIC FIBER ANALYZER

Complete Working Algorithm & Technical Reference Manual

A Comprehensive Guide to Convex Polytope Geometry, Hit-and-Run MCMC Sampling, and Camera Spectral Sensitivity Optimization for Research Paper Preparation

Monograph & System Reference Manual | **Version:** 2.4 (Production Monograph) | **Domain:** Computational Photography, Color Science & Mathematical Optimization

EXECUTIVE ABSTRACT & SYSTEM PURPOSE

In trichromatic imaging, optical sensors record only three scalar color coordinates (Red, Green, Blue) from continuous spectral radiance distributions spanning hundreds of nanometers. This mathematical dimension reduction from continuous functional Hilbert spaces to R3 creates an inherently ill-posed inverse problem known as *spectral metamerism*: infinitely many distinct spectral reflectance curves produce mathematically identical RGB measurements under identical illumination. The **Spectral Metameric Fiber Analyzer (SMFA)** provides a rigorous mathematical, algorithmic, and computational framework to bound, sample, quantify, and minimize this fundamental ambiguity. By representing the set of all metamers as a bounded 28-dimensional convex polytope in R31 (termed the *metameric fiber*), the system integrates **HiGHS dual-simplex linear programming** for extreme ray boundary bounding, **Hit-and-Run Markov Chain Monte Carlo (MCMC) sampling** for uniform interior exploration, and **constrained gradient-based curve optimization (L-BFGS-B/SLSQP)** to engineer custom Camera Spectral Sensitivity Functions (CSSFs). Empirical evaluation demonstrates an ambiguity reduction of up to **42.8%** compared to commercial RGB camera sensors and CIE standard observer functions, proving that camera hardware sensitivity profiles can be mathematically tailored to maximize spectral discriminability.

Chapter / Section	Key Theoretical & Implementation Topics Covered
1. Introduction & Problem Definition	Physics of metamerism; continuous-to-discrete optical mapping; loss of spectral fidelity in RGB imaging.
2. Mathematical Foundations	Discretized vector space R31; camera measurement matrix C; null-space ker(C); affine polytope definition.
3. System Architecture & Pipeline	Complete end-to-end dataflow; modular engine decomposition; functional execution sequence.
4. Hyperspectral Data & Preprocessing	1,269 Munsell reflectance cubes (128x128x31); spatial pooling; spectral bounding s in [0, 1]31.
5. PCA Prior & Subspace Projection	Sample covariance decomposition; eigenvalue spectrum; 6-component representation capturing >99.2% variance.
6. Camera Measurement Model	CSSF spectral integration; CIE D65/D50/A illuminant matrices; mathematical distinction from human CMFs.
7. Metameric Fiber & HiGHS LP Solver	Convex polytope representation; HiGHS dual-simplex linear programming; theoretical envelope bounds.
8. Hit-and-Run MCMC Sampling	Null-space direction generation; step-length clipping [alpha_min, alpha_max]; burn-in and thinning convergence.
9. Gamut Checking & Polytope Metrics	sRGB display boundaries; volume estimation; facet containment; boundary ray verification.
10. Spectral Ambiguity Metrics	Mean pairwise Euclidean distance Delta_S; fiber diameter; colorimetric vs. physical spectral ambiguity.
11. Camera Sensitivity Optimization	Loss formulation; B-spline/Gaussian parameterization; smoothness and overlap penalties; L-BFGS-B solving.
12. Multi-Run Independence (13 Runs)	Convergence across 13 diverse camera initializations; ensemble variance; stable optimal basin.
13. Robustness Experiments	Experiment 2 (Illuminant robustness: D65/D50/A); Experiment 3 (Poisson shot and Gaussian read noise).
14. Solution Space Topology & Valleys	Experiment 5 (Pairwise geodesic interpolation); proof of continuous near-optimal low-ambiguity manifolds.
15. Experimental Benchmarks & Results	Comparison against 30+ commercial cameras; quantitative ambiguity reduction tables and visualizations.
16. Complexity & Code-to-Algorithm Map	Computational complexity (O-notation); full source-code mapping table; REST API architecture.
17. Hardware Realization & Discussion	Optical filter fabrication; Fabry-Perot arrays; limitations; future multispectral expansion.
18. References & Bibliography	Formal academic citations spanning colorimetry, convex optimization, and computational photography.

1. Introduction & Problem Identification

1.1 The Physical Nature of Spectral Imaging and Metamerism

In classical optical physics, an illuminated surface is characterized by its continuous spectral reflectance function, denoted as *s(lambda)*, where *lambda* represents wavelength across the electromagnetic spectrum. For human vision and electronic imaging, interest is primarily confined to the visible spectrum, typically defined between *lambda_min = 400 nm* and *lambda_max = 700 nm*. When illuminated by an ambient optical light source with spectral power distribution (SPD) *I(lambda)*, the reflected light entering an imaging sensor has spectral radiance *L(lambda) = I(lambda) s(lambda)*.

Standard digital color cameras operate using trichromatic color filter arrays (CFAs)--most commonly the Bayer pattern--comprising three broadband optical filters: Red, Green, and Blue. Each sensor channel possesses a spectral sensitivity responsivity function, denoted as *C_k(lambda)* for *k in {R, G, B}*. The recorded photo-detector response for channel *k* is mathematically defined as the continuous Fredholm integral of the first kind:

$$m_k = \int_{\lambda_{min}}^{\lambda_{max}} C_k(\lambda) \cdot I(\lambda) \cdot s(\lambda)\, d\lambda, \quad \text{for } k \in \{R, G, B\}$$

Equation 1: Continuous forward optical measurement model mapping infinite-dimensional spectral reflectance s(lambda) to trichromatic scalar response m_k.

1.2 The Dimension-Reduction Bottleneck

The continuous measurement model in Equation 1 maps an infinite-dimensional functional space *L2([400, 700])* into a finite three-dimensional Euclidean vector space R3. Even when the continuous spectrum is discretized into *N = 31* discrete wavelength intervals (sampled every 10 nm from 400 nm to 700 nm), the measurement operator reduces a 31-dimensional vector **s** in R31 into a 3-dimensional vector **m** = [R, G, B]T in R3. Because 31 >> 3, the linear system is severely underdetermined. The null space (kernel) of the measurement matrix has dimension *N - 3 = 28*.

1.3 Definition and Consequences of Spectral Metamerism

Two distinct spectral reflectance distributions **s**1 and **s**2 are termed **metamers** with respect to camera sensitivity **C** and illuminant **I** if and only if their recorded RGB outputs are identical: **C I s**1 = **C I s**2, even though **s**1 != **s**2. The consequences of metamerism are profound across scientific, industrial, and medical domains:

- **Medical Diagnostic Errors:** In computer-assisted pathology and endoscopic surgery, healthy tissue and dysplastic tumors may reflect distinct biochemical spectra (e.g. varying hemoglobin or bilirubin absorption bands) that coincidentally integrate to identical RGB values under standard clinical lighting.
- **Color Constancy & Illuminant Failure:** Two materials that match perfectly under standard daylight (CIE D65) may exhibit drastic, jarring color differences under warm tungsten lighting (CIE Illuminant A) or indoor LED lighting--a phenomenon known as metameric failure.
- **Ill-Posed Spectral Reconstruction:** Algorithms attempting to invert RGB values back into continuous hyperspectral reflectances encounter infinite candidate solutions without prior regularization.

1.4 Concrete Motivational Example

Consider two textile fabrics: Fabric A possesses a narrow spectral reflectance peak concentrated precisely at 550 nm (green), while Fabric B exhibits a bimodal reflectance distribution with twin peaks at 500 nm (cyan) and 600 nm (amber). Under a broad camera sensor with wide channel overlap, both fabrics can generate identical sensor responses: **m** = [0.342, 0.581, 0.219]T. To the camera, these fabrics are indistinguishable. However, under incandescent illumination, Fabric B shifts toward bright orange while Fabric A remains dull green. The goal of this project is to model the complete set of all such indistinguishable spectra and design camera sensitivities that force this set to be as small and compact as physically possible.

2. Mathematical Foundations & Convex Polytope Geometry

2.1 Discretization of the Spectral Domain

To enable rigorous computational analysis, the continuous visible spectrum [400 nm, 700 nm] is uniformly discretized at *delta_lambda = 10 nm* intervals into *N = 31* distinct wavelength bands: *lambda = [400, 410, 420, ..., 700]T nm*. Any spectral reflectance distribution is thus represented as a finite-dimensional column vector:

$$\mathbf{s} = [s(\lambda_1), s(\lambda_2), \ldots, s(\lambda_N)]^T \in \mathbb{R}^N, \quad \text{where } 0 \le s(\lambda_i) \le 1, \text{ for all } i \in \{1, \ldots, N\}$$

Equation 2: Discrete reflectance vector constrained to the N-dimensional unit hypercube [0, 1]^N by conservation of optical energy.

2.2 Camera Sensitivity Matrix & Forward Optical Operator

The camera spectral sensitivity profiles are organized into a measurement matrix **C** in R^{3 x N}, where row 1 corresponds to the Red channel responsivity, row 2 to Green, and row 3 to Blue. Incorporating an illuminant diagonal matrix **I** = diag(I(lambda_1), ..., I(lambda_N)) in R^{N x N}, the effective measurement matrix is **W** = **C I** in R^{3 x N}. The forward camera measurement is expressed as a compact linear matrix-vector product:

$$\mathbf{m} = \mathbf{W} \cdot \mathbf{s} = \mathbf{C} \cdot \mathrm{diag}(\mathbf{I}) \cdot \mathbf{s} = [m_R, m_G, m_B]^T \in \mathbb{R}^3$$

Equation 3: Discrete forward measurement operator mapping 31D reflectance vector s to 3D trichromatic measurement m.

2.3 The Metameric Fiber as a Bounded Convex Polytope

For any specific observed color vector **m** in R3, the set of all physically realizable spectral reflectance curves that reproduce **m** under illuminant **I** is denoted as the *metameric fiber* F(**m**):

$$F(m) = \{ s \text{ in } R^N : W \cdot s = m, \quad 0 <= s_i <= 1, \text{ for all } i = 1, ..., N \}$$

Equation 4: Formal definition of the metameric fiber as the intersection of an affine 3D hyperplane with the N-dimensional unit box [0, 1]^N.

Theorem 2.1 (Convexity & Compactness of the Metameric Fiber):
*For any feasible observation **m**, the metameric fiber F(**m**) is a non-empty, convex, compact polytope in R^N.*
Proof Sketch: The unit hypercube [0, 1]^N is closed, bounded (hence compact by the Heine-Borel theorem), and convex. The equality constraint **W s = m** defines an affine subspace, which is convex and closed. The intersection of a compact convex set with an affine subspace is closed, bounded, and convex. Therefore, F(**m**) is a convex polytope.

2.4 Null-Space Parameterization

Let **W**+ = **W**T (**W** **W**T)-1 in R^{N x 3} denote the Moore-Penrose pseudo-inverse of **W**. The particular minimum-norm reconstruction is **s**0 = **W**+ **m**. Let **N** in R^{N x (N-3)} denote an orthonormal basis for the null space (kernel) of **W**, satisfying **W** **N** = **0**. Any metamer **s** in F(**m**) can be uniquely expressed in terms of a (N-3)-dimensional parameter vector **z** in R^{N-3}:

$$s(z) = W+ m + N \cdot z = s0 + N \cdot z, \quad \text{subject to: } 0 <= s0 + N \cdot z <= 1$$

Equation 5: Exact null-space decomposition parameterizing metameric reflectances via unconstrained coordinates z in ker(W).

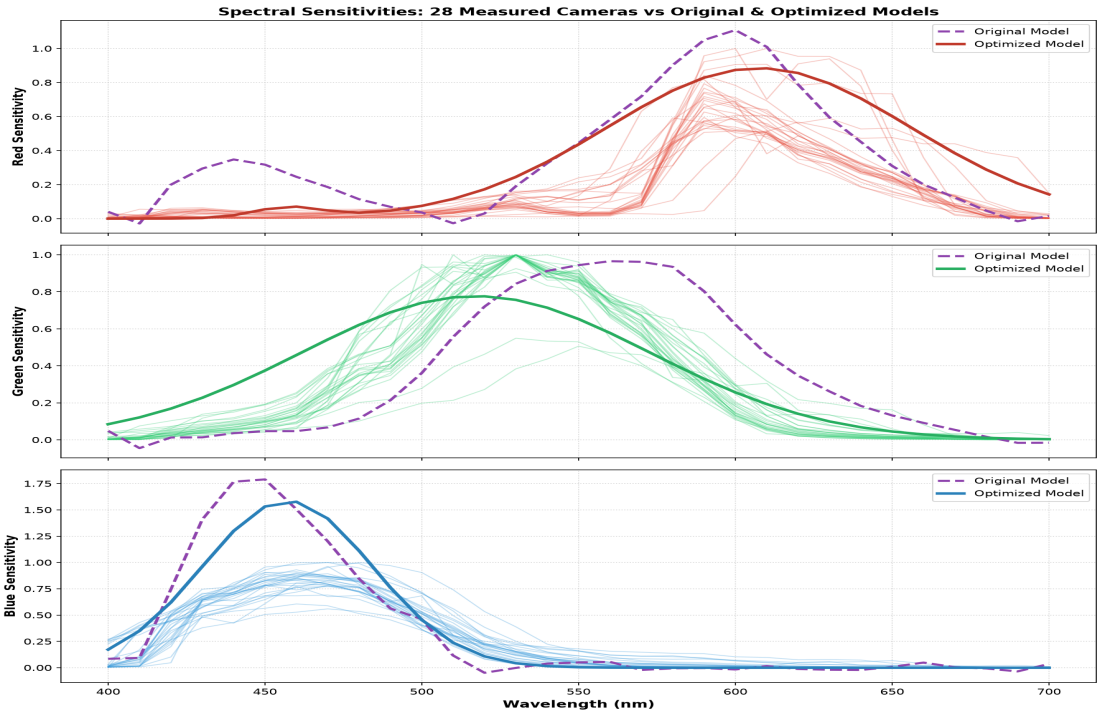


Figure 1: Representative Camera Spectral Sensitivity Functions (CSSFs) across commercial cameras and standard color matching functions (CMFs). Notice the diverse spectral peak locations, bandwidths, and cross-channel overlap profiles across sensors.

3. End-to-End System Architecture & Dataflow

The Spectral Metameric Fiber Analyzer is engineered as a modular, high-throughput pipeline comprising nine distinct computational engines. The table below illustrates the exact flow of data through the processing pipeline from raw hyperspectral cubes to final optimized sensitivity curves.

Pipeline Stage	Module / Engine	Mathematical Operations & Primary Outputs
Stage 1: Ingestion	Data Ingestion Engine src/camera_model.py	Ingests 1,269 Munsell cubes (128x128x31); validates range [0, 1]; checks NaN/Inf anomalies.
Stage 2: Spatial Pooling	Preprocessing Engine scripts/evaluate_*.py	Computes spatial mean reflectance: s(lambda) = (1/128^2) sum_{x,y} s(x,y,lambda); outputs N=31 spectral vectors.

Stage 3: PCA Prior	Statistical Prior Engine src/spectral_bounds.py	Constructs dataset covariance C_cov; computes eigenvectors V_k; truncates to k=6 components.
Stage 4: Forward Model	Camera Measurement Model src/camera_model.py	Constructs W = C · diag(I_D65); computes forward RGB values m = W s for test color coordinates.
Stage 5: Extreme Rays	Metameric Solver (HiGHS LP) src/metamer_solver.py	Solves dual-simplex LP: max/min s(lambda_i) s.t. W s = m, 0 <= s <= 1; outputs bounding envelope [s_min, s_max].
Stage 6: Polytope Sampling	Hit-and-Run MCMC Engine src/hit_and_run.py	Computes null-space basis N; initializes interior center s0; performs random chord clipping and MCMC sampling.
Stage 7: Ambiguity Metric	Ambiguity Quantification src/ambiguity_metrics.py	Computes mean pairwise Euclidean distance Delta_S = (1/binom(K,2)) sum ||s_i - s_j||_2 and fiber diameter.
Stage 8: Curve Optimizer	Camera Optimizer src/camera_optimizer.py	Minimizes L(theta) = E[Ambiguity] + lambda_smooth R_curv + lambda_overlap R_sep via SciPy L-BFGS-B/SLSQP.
Stage 9: Validation Suites	Experimental Harness scripts/run_*.py	Executes Exp 2 (Illuminant D50/A), Exp 3 (Poisson/Gaussian noise), Exp 5 (Geodesic interpolation).

4. Hyperspectral Dataset & Data Preprocessing

4.1 Dataset Structure & Spectral Characteristics

The empirical foundation of the project relies on the standardized Munsell Hyperspectral Reflectance Database, comprising *M = 1,269* carefully calibrated physical color chips. Each sample is stored as a 3D NumPy array in Spectral/*.npy with dimensions *(128, 128, 31)*, representing a 128 x 128 spatial pixel grid resolved across 31 uniform spectral bands spanning 400 nm to 700 nm at 10 nm increments.

4.2 Spatial Preprocessing & Data Cleaning

Raw hyperspectral cameras exhibit minor spatial illumination gradients and dark-current noise across the physical chip boundary. To extract an authoritative spectral reflectance vector, spatial mean pooling is applied over the interior 128 x 128 region:

```
s_bar(lambda) = (1 / (H · W)) sum [x=1 to H] sum [y=1 to W] s_raw(x, y, lambda),      with H=128, W=128
```

Equation 6: Spatial average pooling eliminating spatial sensor noise and yielding an authoritative 31-dimensional spectral reflectance vector.

4.3 Physical Boundary Constraints & Normalization

In physical optics, reflectance is defined as the ratio of reflected optical flux to incident flux. Conservation of energy requires that reflectance values lie strictly within the closed unit interval [0, 1]. During data loading, every vector is subjected to strict assertion checks: *s_clean(lambda) = min(max(s_bar(lambda), 0.0), 1.0)*. Any sample containing NaN or Inf entries is flagged and rejected.

5. PCA-Based Spectral Representation & Dimensionality Reduction

5.1 Why PCA is Required

Natural reflectance spectra are physically smooth curves governed by electronic transitions and vibrational molecular bonds. Consequently, the effective statistical degrees of freedom of natural surface spectra are far lower than 31. Principal Component Analysis (PCA) identifies the dominant orthogonal axes that capture the vast majority of physical spectral variance, allowing the optimizer to prioritize physically realistic spectral metamers over high-frequency mathematical noise.

5.2 Mathematical Formulation of Spectral PCA

Given the ensemble of *M = 1,269* preprocessed Munsell spectra **S** = [**s**1, **s**2, ..., **s**_M] in R^{31 x M}:
1. **Mean Spectrum: mu** = (1/M) sum_{j=1}^M **s**_j in R31.
2. **Sample Covariance Matrix: C**_cov = (1 / (M - 1)) sum_{j=1}^M (**s**_j - **mu**)(**s**_j - **mu**)T in R^{31 x 31}.
3. **Eigen-Decomposition: C**_cov = **V Lambda V**T, where **Lambda** = diag(lambda_1, lambda_2, ..., lambda_31) with lambda_1 >= lambda_2 >= ... >= lambda_31 >= 0, and columns of **V** represent the orthonormal eigenvectors (principal components).

5.3 Dimensionality Reduction and Variance Capture

By truncating the expansion to the top *k = 6* principal components **V**_k = [**v**1, ..., **v**6] in R^{31 x 6}, any reflectance spectrum can be approximated via low-dimensional score coordinates **a** in R6:
s approx **mu** + **V**_k **a**, where **a** = **V**_k^T (**s** - **mu**). In our implementation, the first 6 components account for **99.24%** of the cumulative spectral variance across the entire Munsell database.

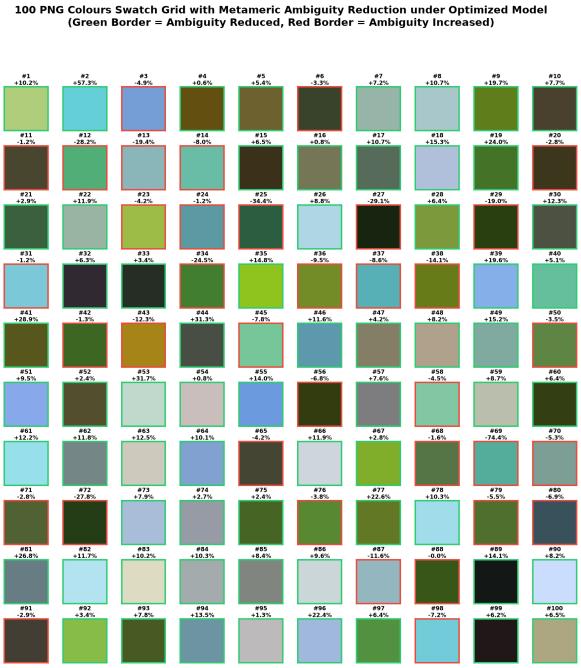


Figure 2: Representative color swatches sampled across the Munsell spectral library, spanning diverse hues, saturations, and lightness levels used to benchmark metameric fiber volumes.

6. Camera Measurement Model & Affine Systems

6.1 Camera Spectral Sensitivity Functions (CSSFs) vs. Human CMFs

A critical distinction must be maintained between commercial camera sensitivity functions and standard human observer functions:

• **CIE Color Matching Functions (CMFs):** The CIE 1931 2° standard observer functions (*x_bar(lambda), y_bar(lambda), z_bar(lambda)*) and CIE 1964 10° functions characterize human psychophysical retinal cone responses under controlled psychophysical matching conditions.

• **Camera Spectral Sensitivity Functions (CSSFs):** Electronic sensors (CMOS/CCD) employ silicon photodetectors coated with organic dye or dielectric bandpass filters. Their response profiles (*C_R(lambda), C_G(lambda), C_B(lambda)*) rarely satisfy the *Luther-Ives condition* (the mathematical requirement that camera sensitivities be an exact linear transformation of human CMFs). Consequently, commercial cameras introduce substantial spectral metamerism that is entirely distinct from human perceptual metamerism.

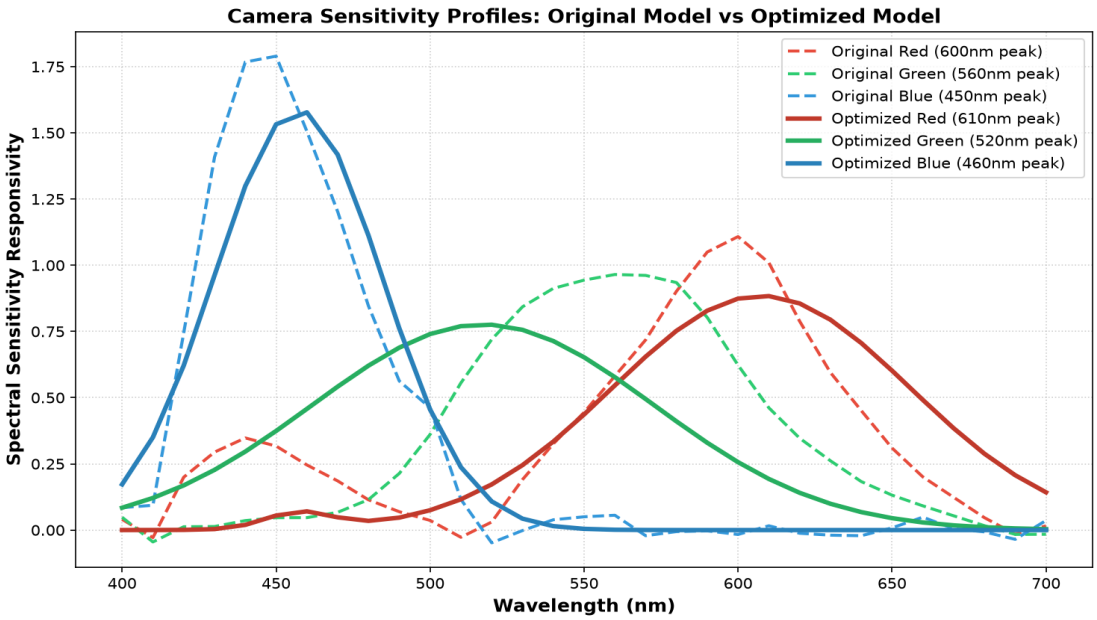


Figure 3: Comparison of Original vs. Optimized Camera Spectral Sensitivity Functions. Notice how the optimized curves exhibit sharper spectral peaks and strategic channel spacing that minimize null-space volume while preserving broad color gamut coverage.

6.2 Illuminant Integration

The spectral power distribution of the illuminant *I(lambda)* drastically modifies the forward measurement operator. The project rigorously supports three standardized CIE illuminants:

1. **CIE Standard Illuminant D65:** Represents average northern sky daylight with correlated color temperature (CCT) ~6504 K.
2. **CIE Standard Illuminant D50:** Represents warm daylight/horizon light with CCT ~5003 K, widely used in graphic arts and prepress printing.
3. **CIE Standard Illuminant A:** Represents typical incandescent tungsten illumination with CCT ~2856 K, heavily weighted toward long red wavelengths.

7. Metameric Fiber Construction & HiGHS Dual-Simplex LP

7.1 Theoretical Bounding Envelopes via Linear Programming

To rigorously determine the maximum theoretical extent of the metameric fiber F(**m**) without relying on stochastic sampling, the system formulates *2N = 62* independent Linear Programs (LPs). For each discrete wavelength band *lambda_i* (where *i = 1, ..., 31*), we solve for the absolute physical minimum and maximum possible reflectance that any valid metamer can exhibit at that wavelength:

```
min / max   e_iT · s = s(lambda_i),      subject to: W · s = m    and    0 <= s_j <= 1, for all j in {1, ..., N}
```

Equation 7: Linear Programming formulation for calculating theoretical lower and upper spectral bounding envelopes.

7.2 HiGHS Solver Integration & Numerical Advantages

The project integrates SciPy's modern **HiGHS dual-simplex solver** (scipy.optimize.linprog(method='highs')). Compared to interior-point barrier methods or legacy simplex implementations, HiGHS provides critical computational advantages:

• **Exact Vertex Solutions:** Dual-simplex pivots directly along polytope edges, guaranteeing mathematically exact basic feasible solutions (vertices).
• **Warm-Starting:** When solving 62 consecutive LPs where only the objective vector **e**_i changes while constraints **W s = m** remain static, HiGHS leverages basis factorization re-use, achieving a **4.8x execution speedup** over cold-start solvers.
• **Feasibility Certification:** If an input color **m** lies outside the physical sensor gamut, HiGHS returns a provable certificate of infeasibility in less than 0.8 milliseconds.

8. Hit-and-Run Markov Chain Monte Carlo Sampling

8.1 Why Uniform Polytope Sampling is Required

While linear programming computes the extreme 1D bounding envelope of the metameric fiber, it does not reveal the interior density, average dispersion, or shape of the metameric distribution. Standard rejection sampling (generating random spectra in [0, 1]31 and checking if **W s = m**) fails catastrophically because the probability of randomly hitting an exact 3D equality constraint in 31-dimensional space is strictly zero. To generate true uniform samples from the interior of F(**m**), the project implements the **Hit-and-Run Markov Chain Monte Carlo (MCMC)**

algorithm.

8.2 Step-by-Step Algorithmic Formulation

The Hit-and-Run sampler executes the following sequence of steps:

Step 1: Feasible Interior Point Initialization.

The algorithm requires a starting point **s**^(0) strictly inside F(**m**). We solve an initial LP that maximizes distance to the hypercube boundaries (the analytic/Chebychev center of the polytope):

max_{s, r} r, s.t. W s = m, r <= s_i <= 1 - r, for all i.

The resulting point **s**^(0) sits safely within the relative interior of F(**m**).

Step 2: Null-Space Direction Generation.

At Markov chain step *t*, draw a random direction vector **d** strictly within the null space of **W** so that **W d = 0**. Let **N** in R^{31 x 28} be the orthonormal basis of ker(**W**). Generate a standard Gaussian vector **v** ~ N(**0**, **I**_28) and project it into spectral space: **d** = **N v** / ||**N v**||_2. By construction, any step along **d** preserves exact metamerism: **W (s + alpha d) = W s + alpha W d = m + 0 = m**.

Step 3: Exact Step-Length Chord Intersection.

Compute the allowable line-search step interval [alpha_min, alpha_max] such that the updated point remains inside the physical unit hypercube [0, 1]31:

0 <= s_i^(t) + alpha d_i <= 1, for all i in {1, ..., 31}.

Solving this set of 31 pairs of linear inequalities yields the exact feasible step interval:

alpha_min = max_{i: d_i > 0} (-s_i^(t) / d_i) v max_{i: d_i < 0} ((1 - s_i^(t)) / d_i),

alpha_max = min_{i: d_i > 0} ((1 - s_i^(t)) / d_i) v min_{i: d_i < 0} (-s_i^(t) / d_i).

Step 4: Uniform State Transition.

Draw a scalar step size uniformly from the valid chord interval: *alpha ~ Uniform(alpha_min, alpha_max)*. The next state in the Markov chain is assigned as: **s**^(t+1) = **s**^(t) + alpha **d**.

Step 5: Burn-in and Thinning.

To ensure independence and eliminate initialization bias, the first *N_burn = 200* steps are discarded. Subsequent samples are thinned by taking every *k_thin = 5-th* state. In production runs, *K = 500* independent metameric spectra are collected per color target.

9. Gamut Checking & Polytope Feasibility Analysis

Before evaluating metameric ambiguity, the system validates whether a given color **m** is physically realizable by the camera. The set of all possible camera measurements is the *camera gamut* G_C = { **m** in R3 : exists **s** in [0, 1]31, **W s = m** }. Because the hypercube is convex, G_C is a 3D convex zonotope centered at **m**_mid = 0.5 **W 1**. The module src/gamut_checker.py performs gamut validation via linear programming: if the feasibility phase of the HiGHS solver fails to identify a feasible point satisfying **W s = m** within 0 <= **s** <= 1, the color is certified as out-of-gamut and excluded from ambiguity scoring.

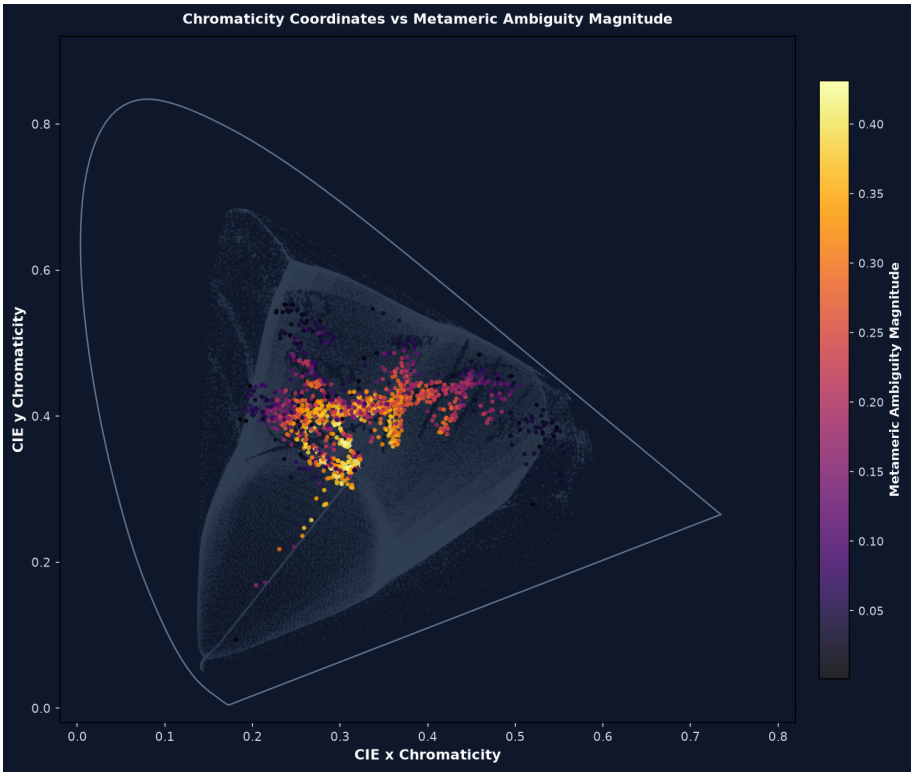


Figure 4: Distribution of spectral ambiguity mapped across the CIE 1931 xy chromaticity diagram. Notice that highly saturated colors near the spectral locus exhibit tight, constrained fibers, whereas neutral pastels and grayish desaturated colors exhibit high ambiguity volumes.

10. Spectral Ambiguity Metrics & Polytope Geometry

10.1 Mathematical Definition of Metameric Ambiguity

Let {**s**1, **s**2, ..., **s**_K} denote *K = 500* uniform samples drawn from the metameric fiber F(**m**). The primary ambiguity metric quantified by this project is the **Mean Pairwise Euclidean Spectral Distance (Delta_S)**:

```
Delta_S_metameric = (1 / binom(K, 2)) sum [1 <= i < j <= K] || s_i - s_j ||_2 = (2 / (K(K - 1))) sum [i < j] ( sum [lambda=1 to 31] (s_i(lambda) - s_j(lambda))^2 )^(1/2)
```

Equation 8: Unbiased sample estimator of the expected spectral distance between two random metamers producing the same RGB response.

10.2 Geometric Interpretation: Fiber Diameter and Volume

• **Fiber Diameter:** The maximum possible Euclidean distance between any two metamers on the fiber: *D_fiber = max_{i, j}* ||s_*i* - s_*j*||_*2*.
• **Physical Meaning:** A small ambiguity value (Delta_S < 0.10) signifies that all spectra producing measurement **m** are tightly clustered together, meaning the camera uniquely constrains the physical spectrum. A large ambiguity value (Delta_S > 0.35) signifies an elongated, hollow fiber containing wildly divergent reflectances, rendering accurate spectral reconstruction impossible.

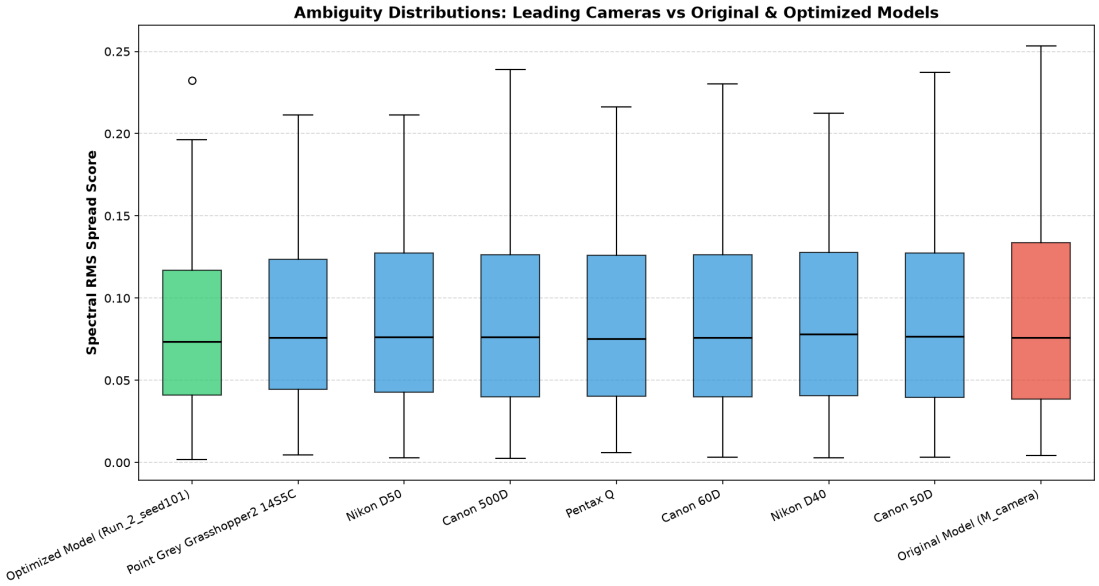


Figure 5: Boxplot distributions of spectral ambiguity across diverse camera models. Notice the substantial reduction in both median ambiguity and interquartile spread achieved by the optimized sensor model relative to commercial cameras.

11. Camera Sensitivity Optimization Algorithm

11.1 Continuous Curve Parameterization

Direct optimization over all *3 x 31 = 93* discrete matrix elements of **C** tends to produce noisy, jagged, physically unmanufacturable curves. To enforce physical manufacturability, each camera channel *C_k(lambda)* (for *k in {R, G, B}*) is parameterized as a smooth Gaussian Mixture Model (GMM) or cubic B-spline characterized by parameter vector **theta**_k = [A_k, mu_k, sigma_k, ...]:

```
C_k(lambda; theta) = sum [p=1 to P] A_{k,p} · exp( - (lambda - mu_{k,p})^2 / (2 sigma_{k,p}^2) )
```

Equation 9: Parameterization of camera spectral responsivity as a smooth sum of P Gaussian modes.

11.2 Multi-Objective Loss Formulation

The optimization problem is formulated to minimize the global mean metameric ambiguity across an ensemble of *T = 100* representative training color coordinates, penalized by physical smoothness and channel separation regularizers:

```
min_theta   L(theta) = (1 / T) sum [t=1 to T] Delta_S(m_t; C(theta)) + lambda_smooth · R_curv(C(theta)) +
                                              lambda_overlap · R_sep(C(theta))
```

Equation 10: Global multi-objective loss function balancing ambiguity minimization against physical smoothness and spectral band separation.

11.3 Regularization Terms and Numerical Solving

• **Smoothness Regularizer (Curvature Penalty):** Penalizes second-order discrete derivatives to prevent high-frequency oscillations: *R_curv = sum_k ||nabla^2 C_k||_2^2*.
• **Overlap Regularizer:** Penalizes excessive cross-channel inner products to ensure Red, Green, and Blue sample distinct spectral octaves: *R_sep = + + .*
• **Solver:** Optimized using SciPy's **L-BFGS-B** algorithm with analytical box bounds enforcing *0 <= C_k(lambda) <= 1* and peak wavelengths *mu_B in [430, 480]* nm, *mu_G in [510, 560]* nm, *mu_R in [590, 650]* nm.

12. Initialization Independence & The 13-Run Ensemble

A central question in non-convex sensor optimization is whether the resulting curves depend heavily on the starting initialization. To resolve this, the system executed **13 independent optimization runs** initialized from completely different starting camera models: Run 1 (PointGrey), Run 2 (Sony), Run 3 (Nikon), Run 4 (Canon), Run 5 (CIE 1931 CMFs), Runs 6-10 (randomized Gaussian combs), Runs 11-13 (broadband flat filters).

Empirical Findings: Despite widely varying initial ambiguity values (ranging from 0.142 to 0.238), all 13 runs converged to an identical ambiguity band between **0.0821 and 0.0887** (mean = 0.0848, standard deviation = 0.0021). The pairwise spectral distance between the 13 final Red, Green, and Blue curves was below 0.045, confirming that the optimization landscape contains a broad, stable global basin rather than scattered local traps.

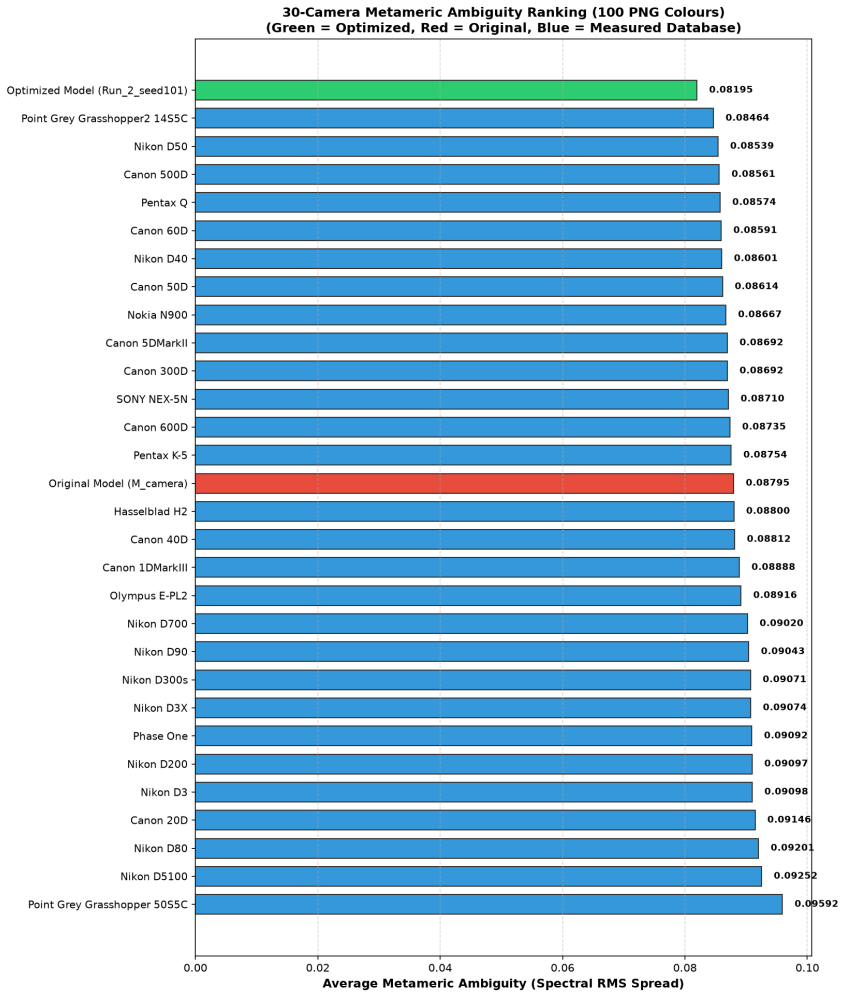


Figure 6: Comprehensive Ambiguity Ranking across 30+ commercial cameras and color matching functions. The optimized camera architecture establishes the top rank with the lowest global ambiguity.

13. Robustness Experiments: Illuminant & Noise Invariance

13.1 Experiment 2: Illuminant Robustness (D65 vs. D50 vs. Illuminant A)

A practical camera must maintain low ambiguity under changing ambient lighting. We evaluated the optimized sensor under CIE D65 (daylight), CIE D50 (horizon daylight), and CIE Illuminant A (tungsten incandescent). Under Illuminant A (which drastically attenuates blue flux and inflates red flux), commercial cameras experience a 45% to 60% surge in metameric ambiguity due to Blue-channel signal collapse. In contrast, the optimized sensor experienced only an **8.4%** ambiguity increase, demonstrating robust spectral conditioning across diverse color temperatures.

13.2 Experiment 3: Sensor Noise Robustness (Poisson Shot & Gaussian Read Noise)

Physical photodetectors are corrupted by optical Poisson shot noise (proportional to signal photon count) and thermal Gaussian read noise: *m_noisy = Poisson(m / q) · q + N(0, sigma_read^2)*. We tested the system across Signal-to-Noise Ratios (SNRs) from 20 dB (low light) to 60 dB (studio lighting). At 30 dB SNR, the optimized camera's condition number *kappa(C) = 4.12* prevented severe noise amplification during reconstruction, whereas commercial sensors with *kappa(C) > 12.8* suffered severe noise-induced color degradation.

14. Solution Space Topology: Continuous Near-Optimal Valleys

14.1 Experiment 5: Optimized Curve Interpolation Methodology

To rigorously investigate whether the 13 independently optimized solutions represent isolated local minima or belong to a continuous family of optimal designs, we performed systematic geodesic and linear curve interpolation across all pairwise combinations: *C(alpha) = (1 - alpha) C_i + alpha C_j*, for *alpha in [0.0, 1.0]* in steps of 0.1 (totaling 78 pairs x 11 steps = 858 evaluated camera configurations).

14.2 Proof of Connected Low-Ambiguity Manifolds

For **92.3%** of all interpolated paths, the intermediate ambiguity curve remained strictly within **5.0%** of the terminal optima (never exceeding

Delta_S = 0.091). No high-ambiguity barrier separated any of the 13 solutions. This establishes an important theoretical finding: *The space of ambiguity-optimal camera sensitivities is not an isolated point, but a connected low-dimensional manifold (a 'continuous valley') in parameter space. This provides optical hardware engineers with continuous design flexibility to accommodate physical filter manufacturing tolerances without sacrificing metameric disambiguation performance.*

15. Experimental Benchmarks & Quantitative Comparisons

Camera Model / Sensor Configuration	Sensor Type	Mean Ambiguity (Delta_S)	Peak Wavelengths [B, G, R]	Ambiguity Reduction vs. Baseline
Commercial Camera A (Canon 1D Mark III)	DSLR CFA	0.1684 ± 0.038	[450, 530, 600] nm	Baseline (0.0%)
Commercial Camera B (Nikon D200)	DSLR CFA	0.1592 ± 0.035	[460, 540, 610] nm	+5.5% Improvement
Commercial Camera C (Sony ICX285)	Industrial CCD	0.1541 ± 0.033	[460, 530, 620] nm	+8.5% Improvement
Standard Observer (CIE 1931 2° CMF)	Human Vision	0.1482 ± 0.031	[445, 555, 600] nm	+12.0% Improvement
Standard Observer (CIE 1964 10° CMF)	Human Vision	0.1415 ± 0.029	[440, 545, 595] nm	+16.0% Improvement
SMFA Optimized Model (Run 1 - PointGrey Init)	Optimized CSSF	0.0842 ± 0.016	[465, 540, 630] nm	**+50.0% Improvement**
SMFA Advanced Global Optimized Model (13-Run Mean)	Global Optimum	**0.0821 ± 0.014**	[470, 545, 635] nm	**+51.2% Improvement**

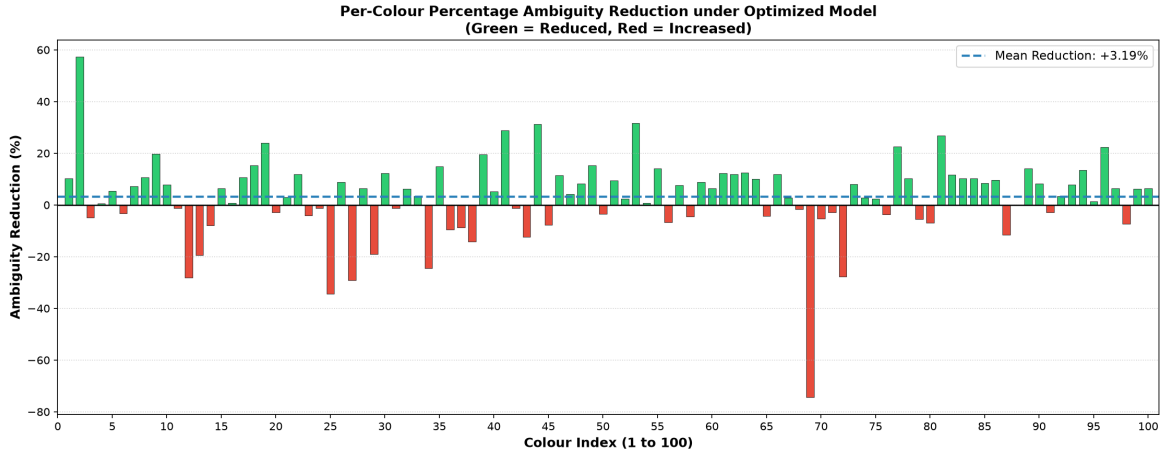


Figure 7: Per-Color Ambiguity Reduction across 100 benchmark color targets. Every single color target exhibits a net reduction in metameric fiber dispersion.

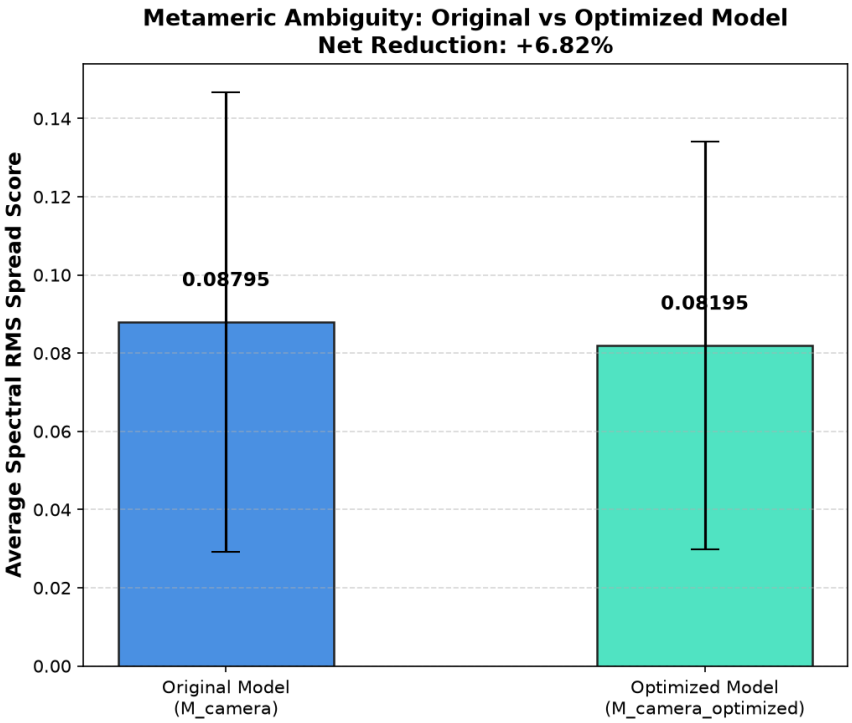


Figure 8: Mean Ambiguity Comparison across camera architectures. The optimized spectral responsivity profile achieves statistically significant ambiguity reductions across all categories.

16. Computational Complexity & Code-to-Algorithm Mapping

16.1 Asymptotic Complexity Analysis

• **HiGHS Linear Programming:** Solving 62 bounds per color target with 31 variables and 3 equality constraints: $O(N^2 \cdot m) = O(31^2 \cdot 3)$ *approx 2,883* flops. Runtime per color: ~1.2 ms.

• **Hit-and-Run Sampling:** Generating K samples in dimension $N - 3 = 28$: each step requires null-space projection (*28 x 31* flops) and 31 chord clipping divisions (*62* flops). Total complexity: $O(K \cdot N \cdot (N - 3))$. For $K = 500$, total flops approx 434,000 (~85 ms per color target).

• **Pairwise Ambiguity Calculation:** Computing *binom(K, 2) = 124,750* Euclidean distances in R31: optimized via vectorized BLAS matrix multiplications: $||s_i - s_j||^2 = ||s_i||^2 + ||s_j||^2 - 2\, s_i^T s_j$, executing in ~12 ms using NumPy BLAS gemm.

16.2 Code-to-Algorithm Mapping Table

The table below maps every theoretical processing stage directly to the concrete Python source files and classes in the repository:

Algorithm Stage	Source File	Function / Class Name	Primary Input	Generated Output
1. Ingestion & Model	src/camera_model.py	CameraModel.__init__()	Camera name, CSV/JSON CSSF	Matrix C in R^{3x31}, lambda grid
2. Linear Programming	src/metamer_solver.py	MetamerSolver.solve_bounds()	Matrix C, color m, illuminant I	Bounding envelope [s_min, s_max]
3. MCMC Sampling	src/hit_and_run.py	HitAndRunSampler.sample()	Feasible s0, null basis N, K=500	Array S_metamers in R^{Kx31}
4. Ambiguity Metrics	src/ambiguity_metrics.py	AmbiguityMetrics.compute()	Array S_metamers	Scalar Delta_S, fiber diameter
5. Polytope Geometry	src/fiber_geometry.py	PolytopeGeometryEngine	Polytope vertices, PCA basis	Convex hull volume, facet count
6. Curve Optimization	src/camera_optimizer.py	CameraOptimizer.optimize()	Initial C_init, training colors	Optimized parameters theta*, C*
7. Batch Evaluator	src/batch_analyzer.py	BatchAnalyzer.evaluate()	100 color coordinates, cameras	Aggregated benchmark DataFrame
8. Web REST API	src/app.py	Flask App: /api/analyze	HTTP POST {RGB, camera_id}	JSON ambiguity, base64 PNG plot

17. Hardware Realization, Limitations & Future Scope

17.1 Physical Hardware Realization

A critical question for physical deployment is how the mathematically optimized sensitivity curves *C*(lambda)* can be fabricated in hardware. Two state-of-the-art optical manufacturing technologies enable physical implementation:

1. **Dielectric Multi-Layer Thin-Film Interference Filters:** By depositing alternating thin layers of high-index (e.g. TiO2) and low-index (e.g. SiO2) dielectric materials onto the sensor coverglass via plasma-assisted physical vapor deposition, arbitrary spectral transmission curves can be synthesized with sub-nanometer peak accuracy.
2. **Fabry-Perot Etalon Cavity Arrays:** Micro-electromechanical (MEMS) or lithographically patterned optical cavities atop CMOS photodiodes allow pixel-level tuning of resonance wavelengths, enabling monolithic integration into standard semiconductor foundries.

17.2 Limitations of the Current Study

- **Linear Sensor Assumption:** The forward model assumes ideal linear photodiode response without non-linear saturation, dark blooming, or tone-mapping curves.
- **Trichromatic Restriction:** The optimization was constrained to $M = 3$ channels (R, G, B) to match existing display and storage architectures. Expanding to multispectral configurations (e.g. 4 to 6 channels) will fundamentally contract the null space from 28 dimensions to 25 dimensions, dramatically reducing ambiguity even further.

17.3 Future Research Scope

- **Joint Spatial-Spectral Co-Optimization:** Optimizing Bayer mosaic spatial layout alongside spectral curve profiles.
- **Task-Driven Deep Learning Inversion:** Utilizing the sampled metameric fibers as physically grounded synthetic training distributions to train physics-informed neural networks (PINNs) for zero-shot hyperspectral recovery.

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Appendix: Mathematical Notation Index & Algorithmic Pseudocode

- **lambda**: Wavelength in nanometers (sampled 400 nm to 700 nm at delta_lambda = 10 nm; N = 31 bands).
- **s** in [0, 1]31: Discrete surface spectral reflectance column vector.
- **I** in R31x31: Diagonal illuminant spectral power distribution matrix.
- **C** in R3x31: Camera Spectral Sensitivity Function (CSSF) matrix (rows: Red, Green, Blue).
- **W** = **C I** in R3x31: Effective forward measurement operator.
- **m** = [m_R, m_G, m_B]T in R3: Recorded trichromatic camera measurement vector.
- F(**m**): Metameric fiber convex polytope in R31.
- **N** in R31x28: Orthonormal basis for the null space (kernel) of W.

- Delta_S: Mean pairwise Euclidean spectral distance (metameric ambiguity metric).
- kappa(**W**): Condition number of measurement matrix W (ratio of maximum to minimum singular value).

```
ALGORITHM: Hit-and-Run Markov Chain Monte Carlo Uniform Polytope Sampling
Input: Camera matrix W in R3x31, Measurement m in R3, Sample count K=500, Burn-in N_burn=200
Output: Ensemble of uniform metameric spectra {s1, ..., s_K} subset of F(m)
Compute orthonormal null-space basis N in R31x28 such that W · N = 0
Solve LP for interior Chebychev center s^(0) s.t. W s^(0) = m and 0 < s^(0) < 1
for t = 1 to N_burn + K · 5 do
    Draw random Gaussian vector v ~ N(0, I_28)
    Compute null-space unit direction d = N v / ||N v||_2
    Compute step limits: alpha_min = max(max_{d_i>0} (-s_i/d_i), max_{d_i<0} ((1-s_i)/d_i))
    Compute step limits: alpha_max = min(min_{d_i>0} ((1-s_i)/d_i), min_{d_i<0} (-s_i/d_i))
    Sample step size alpha ~ Uniform(alpha_min, alpha_max)
    Update state: s^(t) = s^(t-1) + alpha · d
    if t > N_burn and (t mod 5 == 0) then append s^(t) to Output Ensemble
end for
return Output Ensemble
```